

Middlesex University Dubai

Kenya Historia Knowledge
A quiz game website
CST2120 – Web Application and Database
Coursework 1: Game Website

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CST2120_CW1_GAME WEBSITE

Introduction

-The game is called Kenya Historia Knowledge. It will be able to test the user's knowledge on Kenyan History. The player will be presented with four different cards; each card has a different era of Kenyan History. Firstly, the user will roll a dice, and this dice has four sides. Depending on what number the user will land on will determine the card that will be picked.

Project Overview

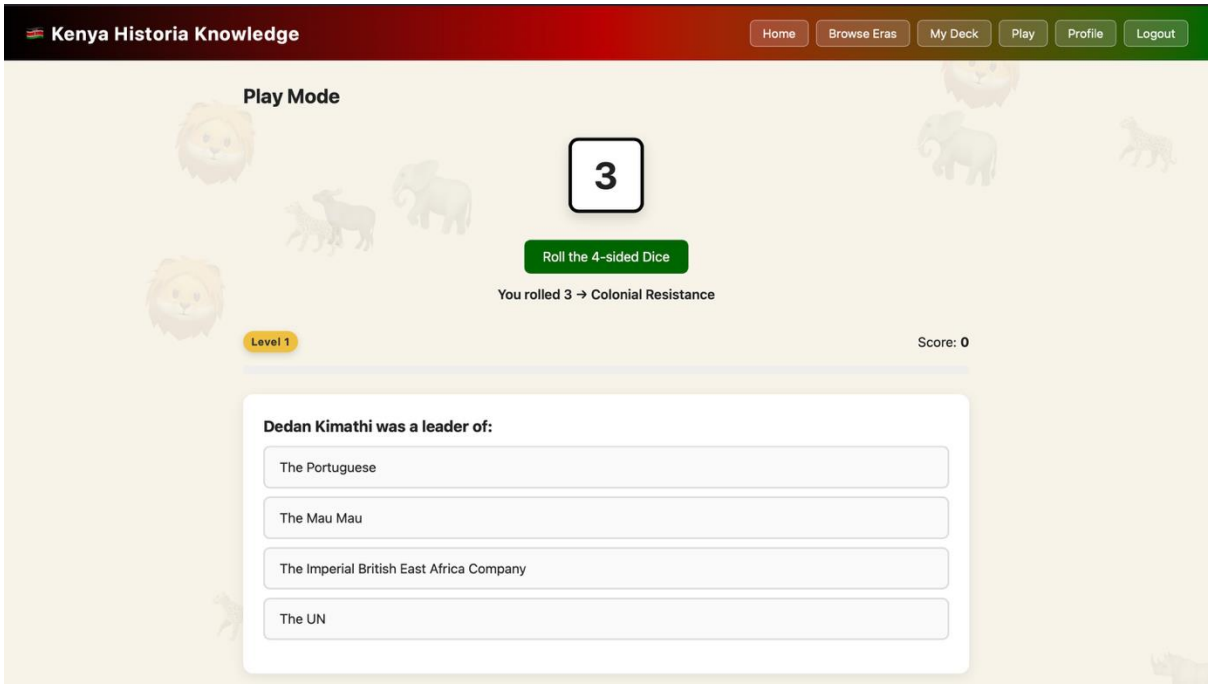
-The aim of the game is to educate the user on Kenyan History. The game will have four era's Pre Colonial, Trade, Independence, Colonial Resistance.
-There are six levels and the questions will become more complex as the game levels up.

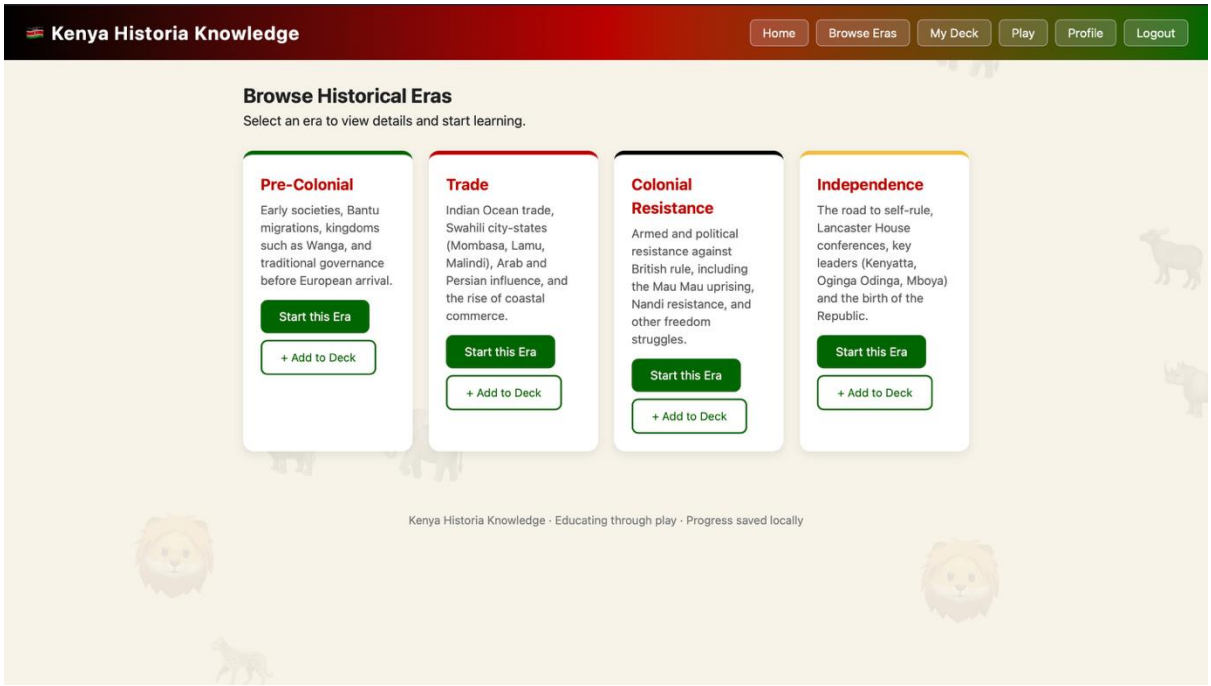
Key features of the game:
User registration and user login
Six progressive levels
Dice rolling
User match history
Top 3 players rankings

Game Mechanics

i) Gameplay

-The player can start the game by rolling the 4-sided dice or browsing through the deck and saving the different eras to their deck. After completing the level, the progress is saved to the match history, if you are a registered user the data will be kept in the local storage but if you're a guest it's in session storage.


Kenya Historia Knowledge
Home
Browse Eras
My Deck
Play
Profile
Logout
Play Mode
3
Roll the 4-sided Dice
You rolled 3 → Colonial Resistance
Level 1
Score: 0
Dedan Kimathi was a leader of:
The Portuguese
The Mau Mau
The Imperial British East Africa Company
The UN

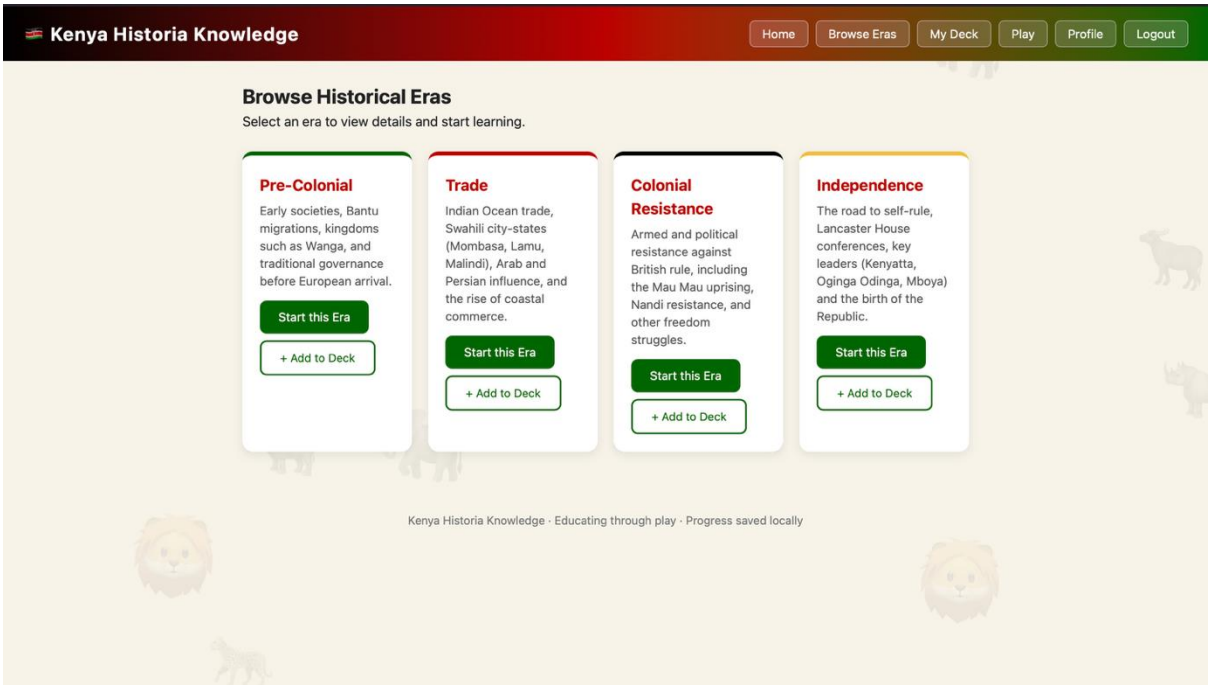

Kenya Historia Knowledge
Home
Browse Eras
My Deck
Play
Profile
Logout
Browse Historical Eras
Select an era to view details and start learning.
Pre-Colonial
Early societies, Bantu migrations, kingdoms such as Wanga, and traditional governance before European arrival.
Start this Era
+ Add to Deck
Trade
Indian Ocean trade, Swahili city-states (Mombasa, Lamu, Malindi), Arab and Persian influence, and the rise of coastal commerce.
Start this Era
+ Add to Deck
Colonial Resistance
Armed and political resistance against British rule, including the Mau Mau uprising, Nandi resistance, and other freedom struggles.
Start this Era
+ Add to Deck
Independence
The road to self-rule, Lancaster House conferences, key leaders (Kenyatta, Oginga Odinga, Mboya) and the birth of the Republic.
Start this Era
+ Add to Deck
Kenya Historia Knowledge · Educating through play · Progress saved locally

ii)Player Control

-The player interacts with the game through the navigation bar, dice rolling button, the next question button, the multiple-choice answer button and the login, register, guest player button.
-This creates a seamless process for the user.

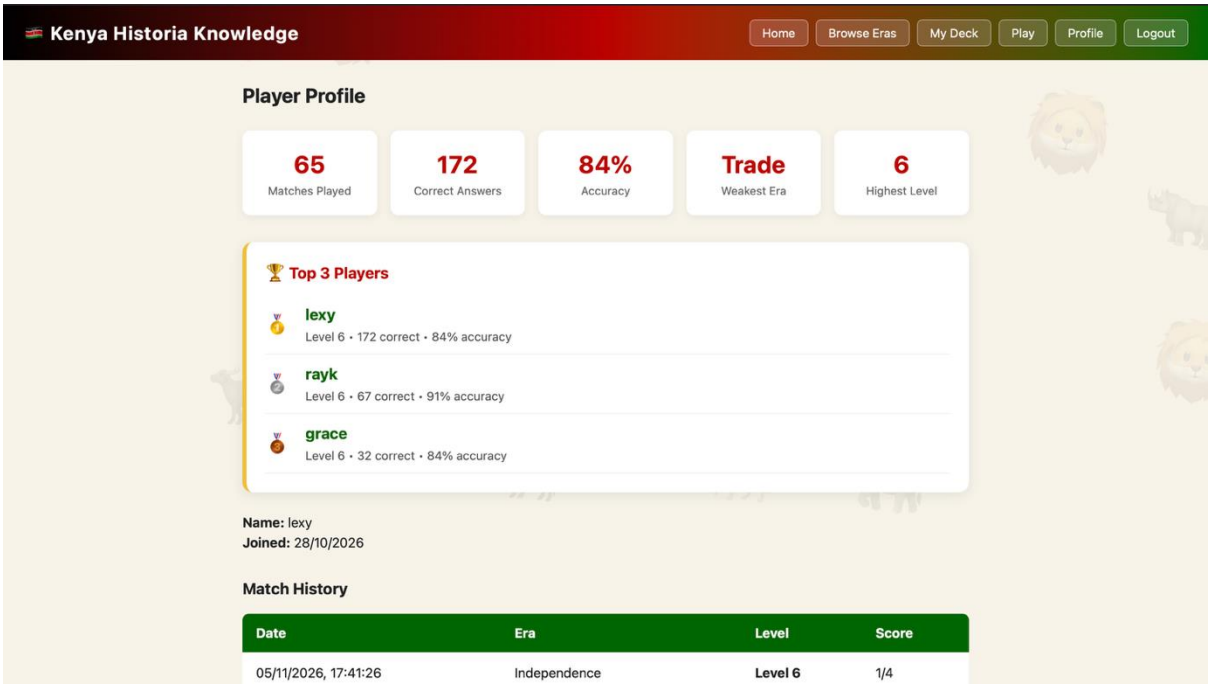
iii)Game Visuals

-The game visual is designed around the Kenyan flag colors which is green, red, black and white.
-The game has different features that are in tune with the Kenyan History like the big five background animation, the card layout of the eras and nav bar.



iv)Advanced Game System

-The players can advance from level one all the way to level six. The user can only progress to the next level after completing the current one.
-The game can record the players score and it's able to depict the user's highest level and weakest era.

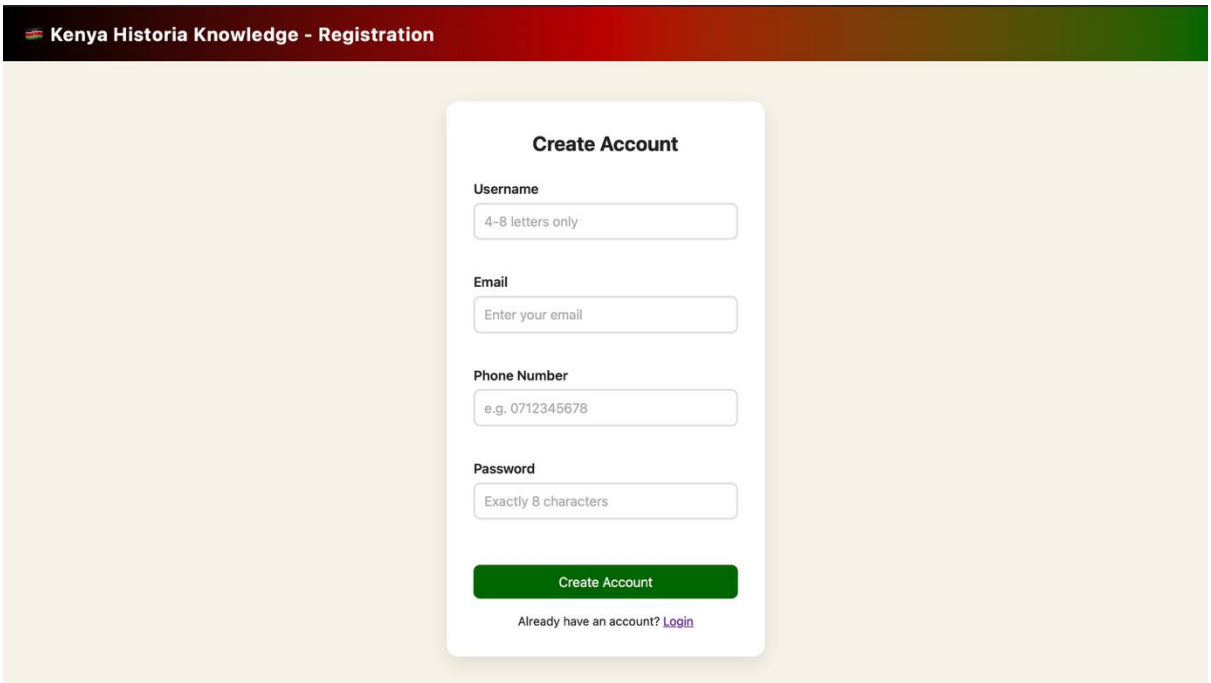


v)Registration System

-New users coming across the game are required to register to track the progress.

-You will need to input the username, email, phone number and password.

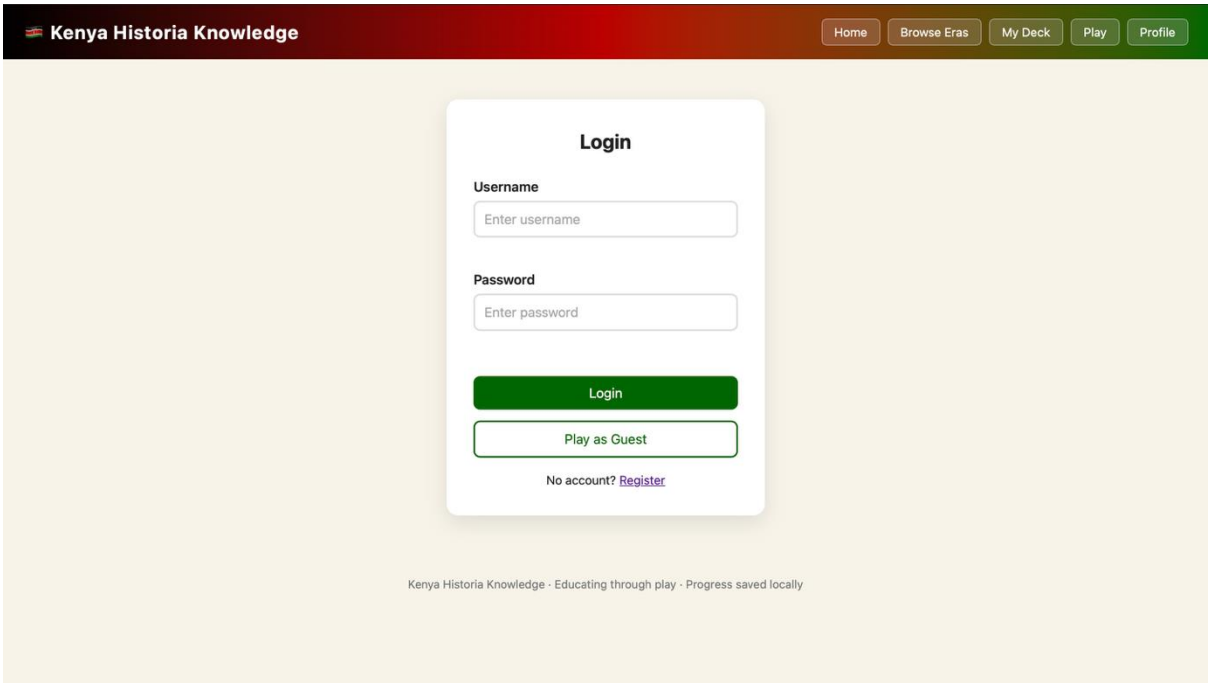
-The system will validate all the inputs to check whether you've put all the necessary requirements.

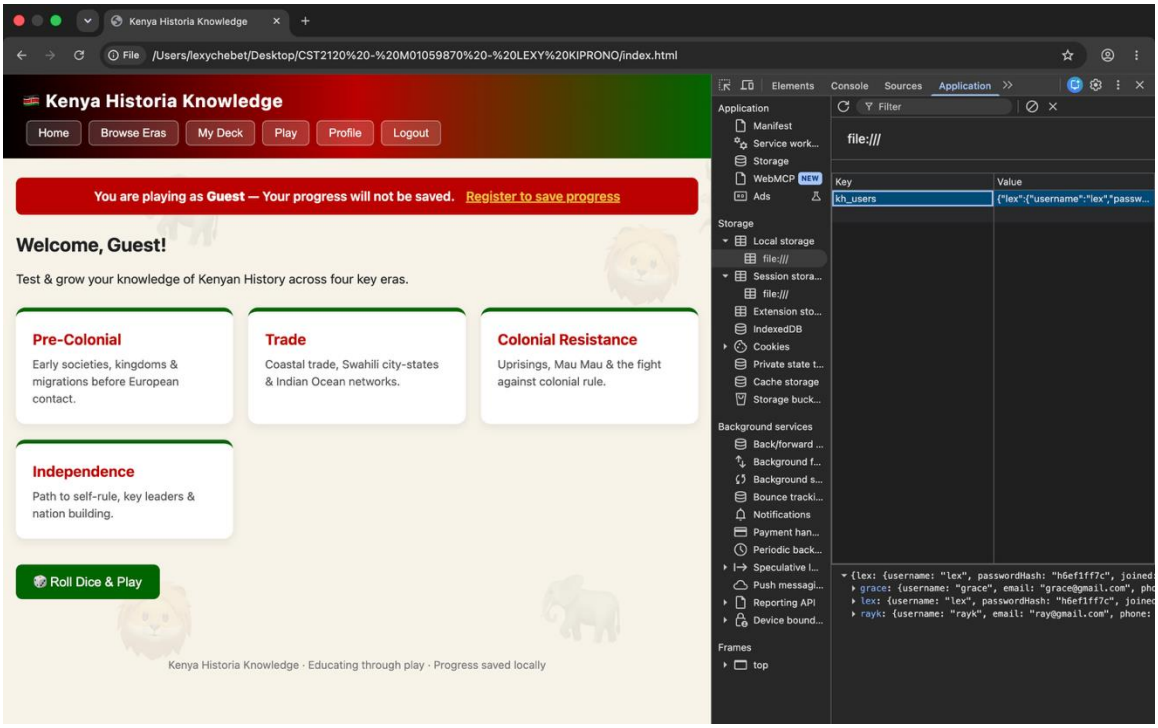


vii)Authentication System

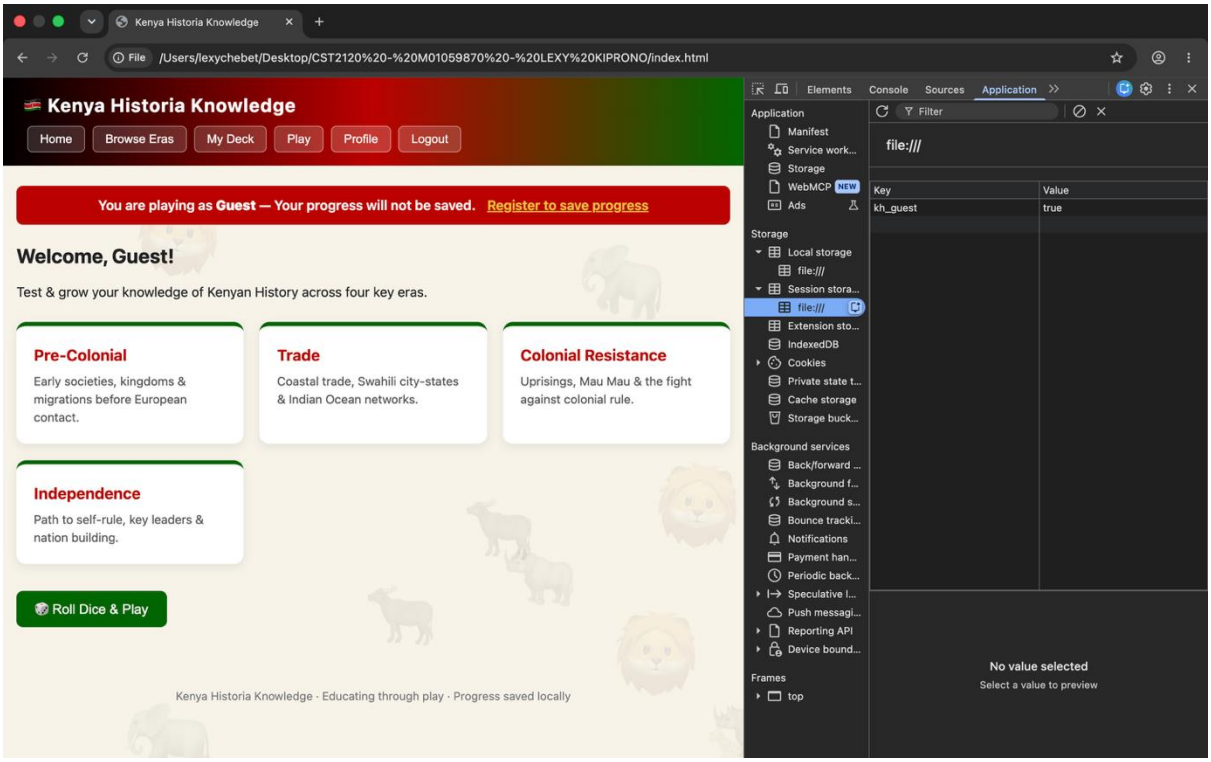
-The registered users can log in using their username and password. The passwords are hashed to maintain user security.

-For the guest players the user's data is temporarily stored in the session storage unlike the registered users the data is saved in the local storage.





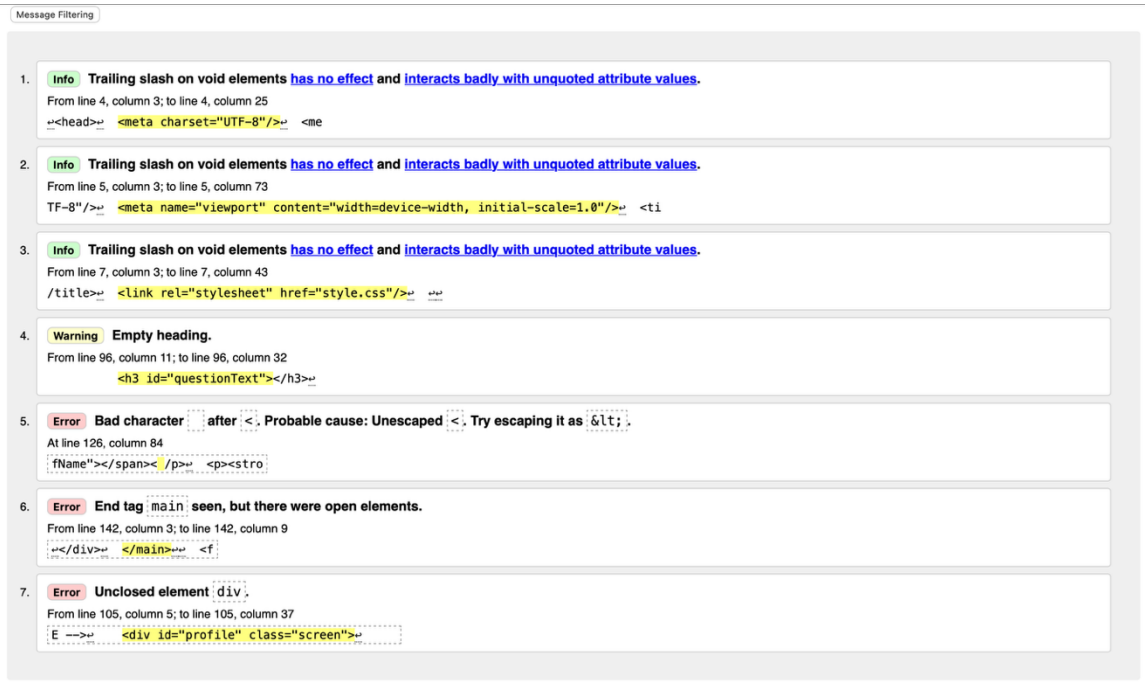
This is the local storage where the registered users are saved.



This is the session storage where the guest data is temporarily stored.

HTML Validation and Error Handling

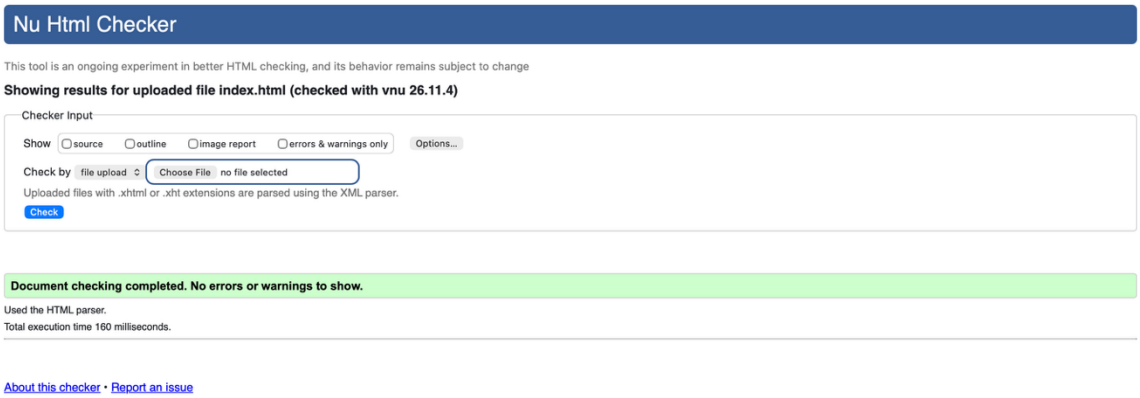
-The HTML structure was tested using W3 validator



Message Filtering

1. Info **Trailing slash on void elements has no effect and interacts badly with unquoted attribute values.**
   From line 4, column 3; to line 4, column 25
   ↵<head>↵ <meta charset="UTF-8"/>↵ <me
2. Info **Trailing slash on void elements has no effect and interacts badly with unquoted attribute values.**
   From line 5, column 3; to line 5, column 73
   TF-8"/>↵ <meta name="viewport" content="width=device-width, initial-scale=1.0"/>↵ <ti
3. Info **Trailing slash on void elements has no effect and interacts badly with unquoted attribute values.**
   From line 7, column 3; to line 7, column 43
   /title>↵ <link rel="stylesheet" href="style.css"/>↵ ↵↵
4. Warning **Empty heading.**
   From line 96, column 11; to line 96, column 32
   <h3 id="questionText"></h3>↵
5. Error **Bad character** ` ` **after** `<`**. Probable cause: Unescaped** `<`**. Try escaping it as** `<`**.**
   At line 126, column 84
   fName">< /p>↵ <p><stro
6. Error **End tag** `main` **seen, but there were open elements.**
   From line 142, column 3; to line 142, column 9
   ↵ ↵ </main>↵↵ <f
7. Error **Unclosed element** `div`**.**
   From line 105, column 5; to line 105, column 37
   E -->↵ ↵

-These are the errors that I had on my main html page and proceeded to fix them.



Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for uploaded file index.html (checked with vnu 26.11.4)

Checker Input

Show ☐ source ☐ outline ☐ image report ☐ errors & warnings only Options...

Check by file upload Choose File no file selected

Uploaded files with .xhtml or .xht extensions are parsed using the XML parser.

Check

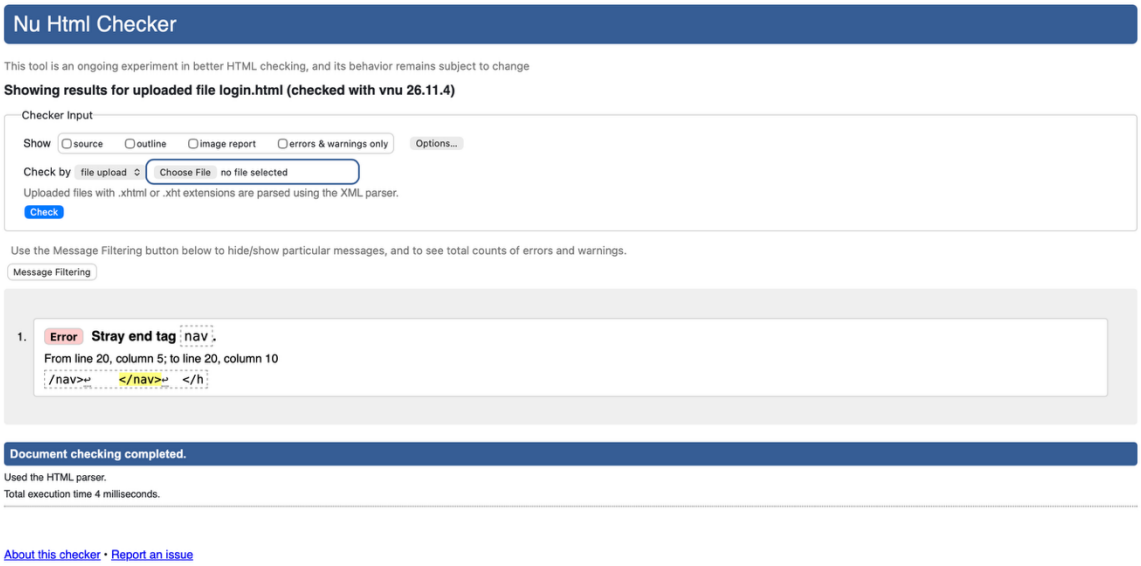
Document checking completed. No errors or warnings to show.

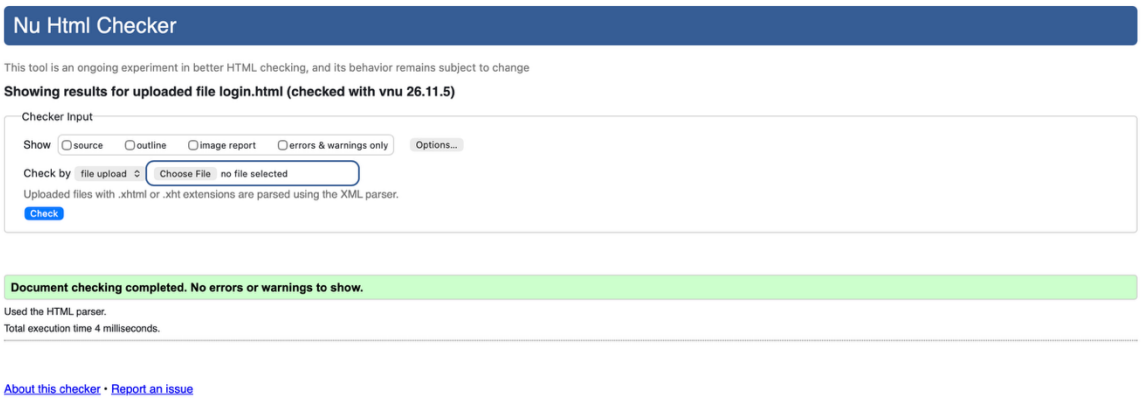
Used the HTML parser.

Total execution time 160 milliseconds.

About this checker • Report an issue

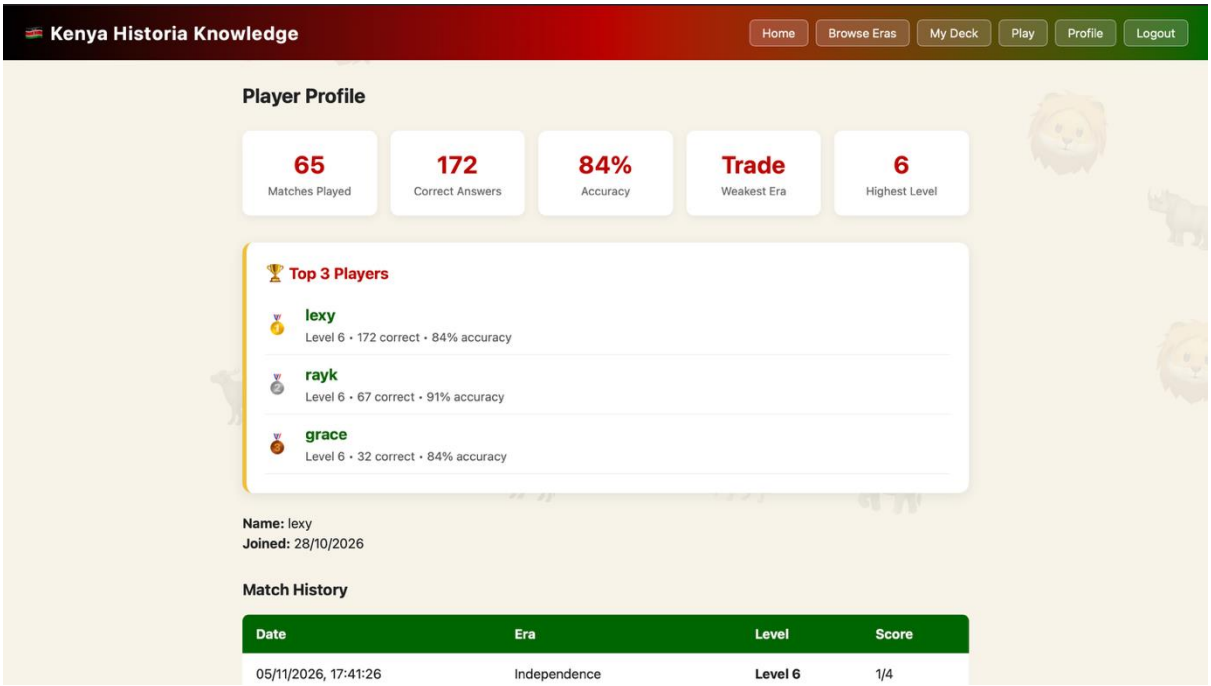
-These are the errors I had on my log in page and proceeded to fix them.





Leaderboard and Rankings

-The game displays the three top players on the user profile page.
-The rankings are based on the total correct answers and the level attained.



Technical Implementation

-The technologies that were implemented are HTML, CSS and JavaScript.
-For the data storage local storage and session storage were used.
-For the sound effects the web API was used.
-Background animation and dice rolling the CSS keyframes were used.
-JavaScript was used for the user profiles, authentication and the gameplay.

File Structure:
index.html – Main game page
login.html – Login page

registration.html – Registration page
index.js – Core game logic
style.css – Styling and animations

Video Link

-This is the YouTube link for the game demonstration
https://youtu.be/ON-C64mquuM

Conclusion

-Kenya Historia Knowledge successfully combines education and entertainment to promote educating about the Kenyan history. The project demonstrates the effective use of front-end web technologies to create an interactive quiz application with user accounts, progression systems, and competitive features.

Future improvements could include:
-Adding more historical questions
-Introducing timed challenges
-Expanding the leaderboard to show more detailed rankings

Overall, the project meets its educational objectives while providing an engaging user experience.

References

HTML tutorial (no date a) *W3Schools*. Available at: https://www.w3schools.com/html/default.asp (Accessed: 07 November 2026).

Class fields and methods : web.dev (no date) *web.dev*. Available at: https://web.dev/learn/javascript/classes/class-fields?continue=https%3A%2F%2Fweb.dev%2Flearn%2Fjavascript%23article-https%3A%2F%2Fweb.dev%2Flearn%2Fjavascript%2Fclasses%2Fclass-fields (Accessed: 07 November 2026).

Artificial Intelligence was used as a point of reference in the refinement of the colour scheme of the Kenya Historia Website.